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Reaching theoretical strengths in nanocrystalline Cu by grain boundary doping

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The yield strength of dilute nc-Cu alloys was investigated using molecular dynamics simulations. Alloying additions that lower grain boundary energy were found to dramatically increase the yield strength of the alloy, with dilute Cu–Nb alloys approaching the theoretical strength of Cu. These findings suggest a new scaling behavior for the onset of plasticity in nanocrystalline materials, one that depends on the product of the specific grain boundary energy and molar fraction of grain boundary atoms, and not simply on grain size alone.

Keywords: Yield strength; Nanocrystalline alloy; Molecular dynamics; Hall–Petch; Copper

The mechanical properties of nanocrystalline (nc) metals have long been of interest, owing to their potential for high strength applications [1–3]. An intriguing, often reported feature of these materials is that they strengthen with decreasing grain size down to very small grain sizes (10–30 nm), but then they reach a maximum in yield strength. Further reduction in grain size results in softening. While this so-called "inverse Hall–Petch" behavior has been discussed for several years [4], the underlying reasons for it have remained largely a matter of conjecture [3]. Part of the reason for this lack of understanding stems from the difficulty in controlling and characterizing the structure and purity of nc-materials, particularly when their grain sizes fall below ~20 nm. Several experimental and simulation studies have indeed pointed out that the strength of nc-materials in this grain-size regime can be influenced by both thermal annealing [5–8] and the addition of solute or impurities [5,6,9–12].

Molecular dynamics (MD) simulations have proved very helpful in providing insight into the strength of nc materials at the atomistic level. Past simulations, which have mostly reported on pure nc samples [13–17], have confirmed the existence of a cross-over from Hall–Petch strengthening to "inverse" Hall–Petch softening. For nc-Cu this occurs at ~10–20 nm in grain size. In previous work, this softening behavior was shown to be largely due to artificially high grain boundary (GB) energies generated in the computer models. In fact, it was demonstrated that when these "as prepared" samples are thermally annealed at high temperatures for several hundreds of picoseconds their strengths dramatically increase and the crossover shifts to much smaller grain sizes [7]. This behavior was explained by showing a direct relationship in nc-Cu between GB relaxation, or energy, and yield strength [8]. The present work extends this idea by using solute atoms to vary the GB energy, rather than by annealing or varying the grain size.

Recent MD simulation studies have indeed begun to consider the effects of solute on strength in nc-metals. Caro et al. [11], for example, reported a small increase in strength of nc-Cu on adding ~2 at.% Fe (i.e., average sample concentration) to the GB. These authors used a combined Monte Carlo (MC)/MD algorithm to globally relax the initial structure [11]. Rajgarhia et al. also showed that a small increase in the strength of nc-Cu of ~10% could be gained by adding up to ~1 at.% Sb to the GB [12]. Unlike the Caro study, the solute in this study was randomly placed in the GB. In a related work, Millett et al. examined the effect of solute on GB sliding in bicrystal samples [18] and reported that solute with a large size mismatch with respect to matrix atoms strongly suppresses sliding. Schäfer et al. showed, moreover, that the state of GB relaxation controls the yield strength in miscible nc-alloys for grain sizes ranging from 5 to 15 nm, and

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that intrinsic material properties, e.g., the stacking fault energy, play a lesser role [19]. While these various results clearly point to an effect of solute on strength in nc-metals, a predictive model remains lacking. Here, a general framework is developed by MD simulation to investigate how both solute addition and thermal relaxation strengthen nc-materials, and for samples in the inverse Hall–Petch regime, it is shown that the yield strength is simply proportional to the product of the specific GB energy (i.e., total GB energy per GB atom) and the molar fraction of GB atoms.

The simulations were performed using dilute Cu alloys as a model system, represented by embedded atom method potentials. The MD code in LAMMPS [20] was employed for all simulations. The samples in this study were created using a Voronoi polyhedra construction, with the samples containing 38 grains and an average grain size of 8 nm. Samples were then relaxed at 300 K for 20 ps to minimize the enthalpy. Solute atoms were placed in these nc-samples using a hybrid MC/MD approach [21], similar to that employed by Caro et al. [11]. The algorithm employs the semi-grand canonical ensemble to locate solute atoms in the sample. Solute atoms are thus introduced into the initially pure nc-Cu structure by changing the atomic identities of individual atoms, based on a given difference in chemical potential. This procedure partitions solute not only between the grain interiors and the GB, but also preferentially among the various GB sites. The method, as employed here, finds the thermodynamic equilibrium distribution of solutes in a constrained microstructure, i.e., precipitation is excluded. Generally, this microstructure is of interest for nc-materials. Details of the algorithm and assumptions are described elsewhere [11,22]. In this work, Nb, Fe and Ag were selected as solutes and placed in nc-Cu samples with the average compositions varying from 0 to ~1.9 at.% for Nb, ~1.5 at.% for Ag, and ~1 at.% for Fe. Interatomic potentials employed for pure Cu and the alloy samples derive from Refs. [23–26], respectively. Partial radial distribution functions for the solutes in GB were calculated to determine whether the solute atoms had precipitated. At higher solute fractions, precipitation was observed in some cases, and these were not used in the present study. In order to understand the significance of placing solute by the MC/MD method (referred to hereafter as MC/MD samples), a second set of nc-Cu–Nb samples was created, with up to ~1.6 at.% Nb, with the solute placed randomly in GB (referred to as random-GB samples). Finally, the yield strengths of the samples were determined from stress–strain curves obtained in compression. Deformation rates of $1 \times 10^9\ \mathrm{s}^{-1}$ were employed. Calculations with lower strain rates of $1 \times 10^8\ \mathrm{s}^{-1}$ showed no qualitative difference. Yield strength is defined here as the stress obtained using a 1% offset in the stress–strain curve, as this represents the approximate strain at which the deformation becomes irreversible in MD simulations (see e.g., Ref. [8]). When the samples do begin to yield, typically at a strain of ~4–5%, the plastic deformation is dominated by GB mechanisms. Using the methods described in Ref. [17], it is found that ~70–80% of the total plastic strain for the 8-nm sample is due to GB sliding and grain rotation rather than to dislocation glide. These findings are consistent with previous work [17,7,8].

Figure 1a shows the atomic configuration of a MC/MD nc-Cu–Nb sample. Here, it is seen that nearly all the Nb is located in the GB. Acceptance rates for atom switching in the GB during construction of the alloys, moreover, are many orders of magnitude higher than those in the grain interiors. This is true for all the alloys studied. The GB energy was calculated according to Weissmüller and co-workers [27–29]:

$$\gamma = \gamma^0 - \Gamma(kT \ln X^L + \Delta H^{\mathrm{seg}}) \tag{1}$$

where γ^0 and γ are GB energy of pure and solute-doped states, respectively, Γ is average coverage of solute in the GB (which is proportional to the solute concentration in the present case), $-k\ln X^L$ is the entropy change of the system on adding GB, and ΔH^{seg} is the segregation enthalpy of the solute atoms. In creating the sample by MC/MD simulation, which was performed at 500 K, the entropy change has only little effect on the free energy. Therefore, the GB energy reduces to

$$\gamma = \gamma^0 - \Gamma \Delta H^{\mathrm{seg}} \tag{2}$$

As the average solute concentration becomes small, the GB energy approaches the GB energy of the pure state. This method of calculation had been used previously [30,31] and, as pointed out in those studies, nc materials are stabilized against grain growth when the GB energy approaches zero. The dependence of GB energy on solute concentration is shown in Figure 1b for the three nc-Cu alloys studied. Here, the specific GB energy is plotted, which is the excess GB energy per GB atom. As discussed in Ref. [31], it is more convenient in MD to calculate the specific GB energy than γ, owing to the difficulty in finding the GB area. The filled symbols in this figure refer to MC/MD alloys, while the open circles refer to random-GB alloys. In each case, the GB energy decreases nearly linearly with average solute concentration, It is observed, moreover, that the addition of ~1.9 at.% Nb (using MC/MD) leads to a reduction in the GB energy to ~0 eV atom^{-1}. Also of note is that the reduction in GB energy in the random-GB samples is significantly smaller than in the MC/MD samples.

Figure 2a shows representative stress–strain curves for pure nc-Cu and nc-Cu doped with Nb. Large enhancements in both the yield and flow strength of solute-doped samples are observed in comparison with the pure sample. This is particularly true for MC/MD samples. Notably, extensive softening is observed in

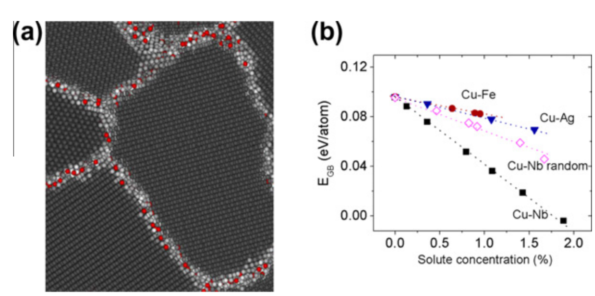

Figure 1. (a) Atomic configuration of an MC/MD Nb-doped Cu nc sample showing that the solute is fully segregated in GB. (b) Specific GB energy as a function of solute concentration in nc-Cu.

the MC/MD samples after the onset of yielding. This softening is presently attributed to the destruction of the well-equilibrated GB structure in these samples. As shown in the inset, the potential energies of the MC/MD samples increase after yielding and approach those for the random-GB samples. The current work, however, focuses only on yield strength.

Figure 2b shows the yield strength in nc-Cu as a function of solute concentration for the three MC/MD alloys investigated. The yield strength, like the specific GB energy, increases nearly linearly with solute concentration up to ~1.2 at.% for the three types of solute. At larger concentration, the strength of the Cu–Nb samples appears to approach saturation. The solute concentration in the GB at this transition point is ~6 at.% for Cu–Nb, and solute–solute interactions are likely to become significant. The yield strength of the sample with 1.2 at.% Nb has increased by ~60% with respect to the pure sample, while for the samples with 1.2 at.% Ag or Fe, the increase is ~30% and ~20%, respectively. These large increases in yield strength are remarkable considering the already high strength of pure nc materials. For example, the strength of defect-free single crystalline Cu deformed in compression along [1 0 0] is ~3.7 GPa [32]. The strength of the nc-Cu–Nb sample thus appears to be saturating very close to the value of this "theoretical" strength of Cu. This observation is analogous to the so-called Borisov model of diffusion [33], for which the activation energy for GB diffusion increases to that of self-diffusion as the GB energy falls to zero.

The correlation between yield strength and specific GB energy, suggested by Figures 1b and 2b, is shown explicitly in Figure 3a, where the yield strengths for all samples are plotted as a function of their GB energy. Nearly all the data for the MC/MD samples lie on a single curve, with only the samples with high concentrations of Nb placed randomly in GB deviating from this behavior. Analysis of these samples shows, however, that these deviations may derive from an overestimation of their GB energies. As shown in the inset in Figure 3a, the fraction of Nb solute without Nb nearest neighbors is larger for the samples created by MC/MD than by random placement, with the difference increasing with solute concentration. This indicates that the MC/MD method better spaces the Nb solute in GB. Since the calculation of GB energy assumes isolated solute in GB, i.e., Henrian behavior, the values of GB energies for the random-GB alloys and those for the high-concentration MC/MD alloys, may be too low. A similar effect (not shown here) was observed for Fe solute.

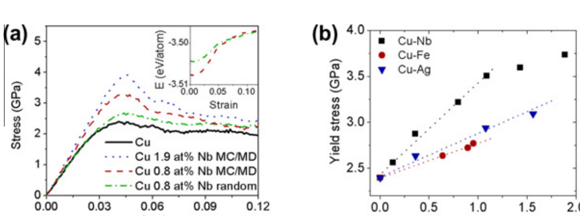

Figure 2. (a) Stress–strain curve of pure and Nb-doped nc-Cu at constant strain rate $1 \times 10^9\ s^{-1}$. Potential energy as a function of strain of Cu 0.8 at.% Nb MC/MD and random–GB samples is shown in inset. (b) Yield stress of MC/MD nc-Cu alloys as a function of solute concentration at constant strain rate $1 \times 10^9\ s^{-1}$.

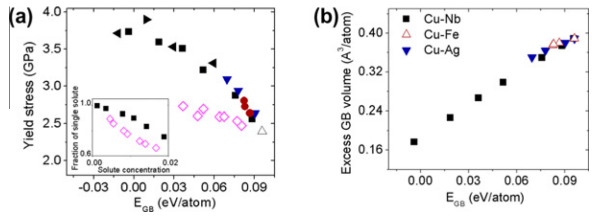

Figure 3. (a) Yield strength as a function of specific GB energy of solute-doped nc-Cu at a strain rate of $1 \times 10^9\ s^{-1}$: pure Cu (Δ), Cu–Nb (■), Cu–Nb random–GB (◊), Cu–Ag (▼), Cu–Fe (●). All samples are 8 nm in diameter except (◄) 4 nm and (►) 15 nm Cu–Nb, which are normalized by N_{GB}/N_{total} value (see text). (b) Linear relationship between specific GB energy and specific GB volume of nc samples.

The correlation between yield strength and GB energy is next explained using the simple model introduced [7] to explain the dependence of flow stress on grain size and GB energy. In that model, the flow stress was written in the form

$$\frac{1}{\sigma} = \frac{1}{k_1 + k_2/d^{1/2}}\left(1 - \frac{N_{GB}}{N_{total}}\right) + k_3 \cdot E_{GB} \cdot \left(\frac{N_{GB}}{N_{total}}\right) \qquad (3)$$

where σ is the flow stress, d is grain size, k_1, k_2 and k_3 are fitting parameters, E_{GB} is the specific GB energy, and N_{GB} and N_{total} are the number of atoms in GB and the total number of atoms, respectively. The two terms on the right-hand side of Eq. (3) represent a competition between traditional Hall–Petch hardening and GB softening (sliding). For samples with a large ratio, N_{GB}/N_{total} (small grain size), the second term dominates, while the first term dominates for samples with relatively few GB atoms (large grain size). Eq. (3) was shown to fit well to a large data set obtained for different grain sizes and thermal annealing times [7]. At small strains of ~4–5%, GB sliding is known to predominate, and the second term is far more important [8,17,34]. The stress in Eq. (3) refers in the present work to the yield stress, i.e., small-strain regime, and consequently the inverse yield stress should scale almost linearly with $E_{GB} \cdot (N_{GB}/N_{total})$ where, for pure samples, E_{GB} is assumed constant (~0.085 eV atom^{-1}). As shown in Ref. [8], N_{GB} decreased during thermal annealing due to GB relaxation, and not to a change in grain size. Eq. (3) then requires only a single fitting parameter (k_3), and the yield strength does not depend simply on grain size.

In contrast to previous work using thermal annealing, N_{GB}/N_{total} is now left nearly constant (~0.25), and E_{GB} is systematically varied by injecting solute into the GB. N_{GB}/N_{total} varies by <~10% in the different samples, since the solute concentration is small. All samples have the same grain size and are annealed for the same amount of time. Figure 3a demonstrates that the predicted scaling of yield strength with specific GB energy, E_{GB}, remains valid for these alloys. As an additional test of this model, nc-Cu–Nb was created with two other grain sizes, 4 and 15 nm. Yield stresses as a function of E_{GB} of these samples doped with different amounts of Nb solute by the MC/MD method are also plotted in Figure 3a. The E_{GB} for these samples has been scaled, however, by N_{GB}/N_{total} for comparison with the 8 nm samples. These data fit very well with the trend of the 8 nm samples. Simulations at a lower strain

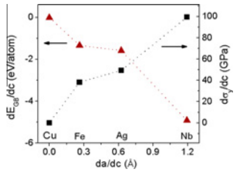

Figure 4. Linear relationships between lattice mismatch of solute and matrix atoms vs GB energy and yield strength of nc samples.

rate (1×10^8 s^{-1}) showed the same trend, although the absolute value of the yield strength decreased by ~10%.

It is noted that recent results reported by Schäfer et al. reveal that the strength of nc-PdAu alloys [19] scale with GB volume. As discussed above, the yield stress in pure nc-Cu scales with N_{GB}, and thus it too scales directly with GB volume. It is now considered whether this scaling also holds for the Cu alloys. For the alloys, however, N_{GB} is no longer a valid measure of GB volume, since the different solutes have different atomic volumes. The GB volume is thus calculated directly as the difference in volume of the nc-sample and a single crystalline sample with the same number of atoms. Figure 3b shows for the samples investigated that the specific GB energy scales linearly with specific GB volume. The linear relationship between these two quantities implies that the yield strength can also be described in terms of the GB volume.

Lastly, this paper comments on the efficacy of each type of solute for increasing the yield strength. Plotted in Figure 4 are the decrement of GB energy (dE_{GB}/dc) and the increment of yield strength ($d\sigma_y/dc$) as a function of lattice constant increment (da/dc) in single-crystalline Cu, i.e., lattice mismatch. The nearly linear trend shows that strength increases nearly proportionately to the atomic size mismatch. This finding, that strength increases with increasing atomic volume of the solute, agrees with the results of Ref. [18], where Lennard Jones potentials were employed.

In summary, MD simulation was used to study yield strength of solute-doped nc-Cu samples. The primary results are that (i) GB energy decreases as solute concentration increases, and (ii) the yield strength scales with GB energy. The study revealed, moreover, that the GB energy depends on the type of solute in the GB and whether the GB are suitably relaxed. Solutes with a large size mismatch are more effective in reducing GB energies. The results suggest a new scaling behavior for the onset of plasticity in nc materials, one that is controlled not by the grain size alone, but by a combination of both the molar fraction of GB atoms and the degree of GB relaxation, as measured by the specific GB energy of the sample.

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